PGCP-AC August 26
Database Assignment No-3

1. Write a query to calculate the total salary of all employees.

```
mysql> select * from Employee;
+-------+-------+-----------------+--------+------------+--------------+
| empId | Name  | emailId         | salary | Department | Joining_Date |
+-------+-------+-----------------+--------+------------+--------------+
|     2 | Ram   | ram@gmail.com   |  60500 | IT         | 2026-10-04   |
|     3 | Priya | priya@gmail.com |  60000 | IT         | 2026-10-06   |
|     4 | Neha  | neha@gmail.com  |  70000 | Comp       | 2026-09-06   |
|     5 | Nakul | nakul@gmail.com |  75000 | Comp       | 2026-07-06   |
|     6 | Sai   | sai@gmail.com   |  74800 | AI         | 2026-07-20   |
|   101 | Rahul | rahul@gmail.com |  63800 | AI         | 2026-09-18   |
+-------+-------+-----------------+--------+------------+--------------+
6 rows in set (0.00 sec)


mysql> select Sum(salary) as "Total Salary" from Employee;
+--------------+
| Total Salary |
+--------------+
|       509100 |
+--------------+
1 row in set (0.00 sec)
```

2. Write a query to find the average salary of employees in each department using GROUP BY.

mysql> select department, avg(salary) from Employee group by department;

+------------+-------------+

| department | avg(salary) |

+------------+-------------+

| IT | 60250.0000 |

| Comp | 72500.0000 |

| AI | 69300.0000 |

+------------+-------------+

3 rows in set (0.00 sec)

3. Write a query to count the total number of employees in each department.

mysql> select department, count(department) from Employee group by department;

+------------+--------------------+

| department | count(department) |

+------------+--------------------+

| IT | 2 |

| Comp | 2 |

| AI | 2 |

+------------+--------------------+

3 rows in set (0.00 sec)

4. Write a query to display departments having more than 5 employees using HAVING clause.

mysql> select department, count(*) from Employee group by department having count(*)>5;

Empty set (0.00 sec)

mysql> select department, count(*) from Employee group by department having count(*)<5;

+------------+----------+

| department | count(*) |

+------------+----------+

| IT | 2 |

| Comp | 2 |

| AI | 2 |

| Mech | 2 |

+------------+----------+

4 rows in set (0.00 sec)

5. Write a query to list distinct department locations from the Dept table.

mysql> select distinct location from Employee;

+----------+

| location |

+----------+

| Kharghar |

| Pune |

| Juhu |

+----------+

3 rows in set (0.00 sec)

6. Write a query to display the highest salary among all employees. Write a query to display all total number of employee in table.

mysql> select max(salary) from Employee;

+-------------+

| max(salary) |

+-------------+

| 75000 |

```
+-------------+
```

1 row in set (0.00 sec)

```
mysql> select count(*) from Employee;
+----------+
| count(*) |
+----------+
|       6 |
+----------+
```

1 row in set (0.00 sec)

7. Write a query to find employees whose name starts with 'A' using LIKE operator.

```
mysql> select * from Employee where Name like 'A%';
+-------+--------+-----------------+--------+------------+--------------+-------------+
| empId | Name   | emailId         | salary | Department | Joining_Date | location    |
+-------+--------+-----------------+--------+------------+--------------+-------------+
|   102 | Anjali | anjali@gmail.com |  50000 | Mech       | 2026-06-02   | Ahilyanagar |
+-------+--------+-----------------+--------+------------+--------------+-------------+
```

1 row in set (0.00 sec)

8. Write a query to find employees whose name ends with 'n' using LIKE operator.

```
mysql> select * from Employee where Name like '%n';
+-------+-------+-----------------+--------+------------+--------------+-------------+
| empId | Name  | emailId         | salary | Department | Joining_Date | location    |
+-------+-------+-----------------+--------+------------+--------------+-------------+
|   103 | Kiran | kiran@gmail.com |  55000 | Mech       | 2026-07-02   | Ahilyanagar |
+-------+-------+-----------------+--------+------------+--------------+-------------+
```

1 row in set (0.00 sec)

9. Write a query to find employees whose name contains 'ra' using LIKE operator.

```
mysql> select * from Employee where Name like '%ra%';
+-------+-------+-----------------+--------+------------+--------------+-------------+
| empId | Name  | emailId         | salary | Department | Joining_Date | location    |
+-------+-------+-----------------+--------+------------+--------------+-------------+
|     2 | Ram   | ram@gmail.com   |  60500 | IT         | 2026-10-04   | Kharghar    |
|   101 | Rahul | rahul@gmail.com |  63800 | AI         | 2026-09-18   | Juhu        |
|   103 | Kiran | kiran@gmail.com |  55000 | Mech       | 2026-07-02   | Ahilyanagar |
+-------+-------+-----------------+--------+------------+--------------+-------------+
3 rows in set (0.00 sec)
```

10. Write a query to display all employees sorted by their Salary in descending order.

```
mysql> select * from Employee order by salary desc ;
+-------+--------+-----------------+--------+------------+--------------+-------------+
| empId | Name   | emailId         | salary | Department | Joining_Date | location    |
+-------+--------+-----------------+--------+------------+--------------+-------------+
|     5 | Nakul  | nakul@gmail.com |  75000 | Comp       | 2026-07-06   | Pune        |
|     6 | Sai    | sai@gmail.com   |  74800 | AI         | 2026-07-20   | Juhu        |
|     4 | Neha   | neha@gmail.com  |  70000 | Comp       | 2026-09-06   | Pune        |
|   101 | Rahul  | rahul@gmail.com |  63800 | AI         | 2026-09-18   | Juhu        |
|     2 | Ram    | ram@gmail.com   |  60500 | IT         | 2026-10-04   | Kharghar    |
|     3 | Priya  | priya@gmail.com |  60000 | IT         | 2026-10-06   | Kharghar    |
|   103 | Kiran  | kiran@gmail.com |  55000 | Mech       | 2026-07-02   | Ahilyanagar |
|   102 | Anjali | anjali@gmail.com |  50000 | Mech       | 2026-06-02   | Ahilyanagar |
+-------+--------+-----------------+--------+------------+--------------+-------------+
8 rows in set (0.00 sec)
```

11. Write a query to display all employees sorted by DeptID ascending and then Salary descending.

mysql> select * from Employee order by department asc, salary desc ;

```
+-------+--------+-----------------+--------+------------+--------------+-------------+
| empId | Name   | emailId         | salary | Department | Joining_Date | location    |
+-------+--------+-----------------+--------+------------+--------------+-------------+
|     6 | Sai    | sai@gmail.com   |  74800 | AI         | 2026-07-20   | Juhu        |
|   101 | Rahul  | rahul@gmail.com |  63800 | AI         | 2026-09-18   | Juhu        |
|     5 | Nakul  | nakul@gmail.com |  75000 | Comp       | 2026-07-06   | Pune        |
|     4 | Neha   | neha@gmail.com  |  70000 | Comp       | 2026-09-06   | Pune        |
|     2 | Ram    | ram@gmail.com   |  60500 | IT         | 2026-10-04   | Kharghar    |
|     3 | Priya  | priya@gmail.com |  60000 | IT         | 2026-10-06   | Kharghar    |
|   103 | Kiran  | kiran@gmail.com |  55000 | Mech       | 2026-07-02   | Ahilyanagar |
|   102 | Anjali | anjali@gmail.com|  50000 | Mech       | 2026-06-02   | Ahilyanagar |
+-------+--------+-----------------+--------+------------+--------------+-------------+
```

8 rows in set (0.00 sec)

12. Write a query to find employees whose salary is between 30,000 and 60,000.

mysql> select * from employee where salary between 30000 and 60000;

```
+-------+--------+-----------------+--------+------------+--------------+-------------+
| empId | Name   | emailId         | salary | Department | Joining_Date | location    |
+-------+--------+-----------------+--------+------------+--------------+-------------+
|     3 | Priya  | priya@gmail.com |  60000 | IT         | 2026-10-06   | Kharghar    |
|   102 | Anjali | anjali@gmail.com|  50000 | Mech       | 2026-06-02   | Ahilyanagar |
|   103 | Kiran  | kiran@gmail.com |  55000 | Mech       | 2026-07-02   | Ahilyanagar |
+-------+--------+-----------------+--------+------------+--------------+-------------+
```

3 rows in set (0.00 sec)

13. Write a query to display all employees whose DeptID is in (10, 20, 30).

```
mysql> select * from Employee where DeptId in (10,20,30);
+-------+-------+-----------------+--------+------------+--------------+----------+--------+
| empId | Name  | emailId         | salary | Department | Joining_Date | location | DeptId |
+-------+-------+-----------------+--------+------------+--------------+----------+--------+
|     2 | Ram   | ram@gmail.com   |  60500 | IT         | 2026-10-04   | Kharghar | 10     |
|     3 | Priya | priya@gmail.com |  60000 | IT         | 2026-10-06   | Kharghar | 10     |
|     4 | Neha  | neha@gmail.com  |  70000 | Comp       | 2026-09-06   | Pune     | 20     |
|     5 | Nakul | nakul@gmail.com |  75000 | Comp       | 2026-07-06   | Pune     | 20     |
|     6 | Sai   | sai@gmail.com   |  74800 | AI         | 2026-07-20   | Juhu     | 30     |
|   101 | Rahul | rahul@gmail.com |  63800 | AI         | 2026-09-18   | Juhu     | 30     |
+-------+-------+-----------------+--------+------------+--------------+----------+--------+
6 rows in set (0.00 sec)
```

14. Write a query to display Min salary of employee.

```
mysql> select min(salary) as "Min salary" from Employee;
+------------+
| Min salary |
+------------+
|      50000 |
+------------+
1 row in set (0.00 sec)
```

15. Write a query to display employees whose JoiningDate is between '2020-01-01' and '2021-12-31'.

```
mysql> select * from employee where joining_date between '2020-01-01' and '2021-12-31';
+-------+-------+-----------------+--------+------------+--------------+----------+--------+
| empId | Name  | emailId         | salary | Department | Joining_Date | location | DeptId |
+-------+-------+-----------------+--------+------------+--------------+----------+--------+
```

```
+-------+-------+-----------------+--------+------------+--------------+----------+--------+
|    2 | Ram   | ram@gmail.com   |  60500 | IT         | 2021-06-04   | Kharghar | 10     |
|    3 | Priya | priya@gmail.com |  60000 | IT         | 2021-07-20   | Kharghar | 10     |
+-------+-------+-----------------+--------+------------+--------------+----------+--------+
2 rows in set (0.00 sec)
```

16. Write a query to display employees whose Salary is NULL.

```
mysql> select * from employee where salary IS NULL;
Empty set (0.00 sec)
```

17. Write a query to display employees whose Salary is NOT NULL.

```
mysql> select * from employee where salary IS NOT NULL;
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
| empId | Name   | emailId         | salary | Department | Joining_Date | location    | DeptId |
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
|    2 | Ram    | ram@gmail.com   |  60500 | IT         | 2021-06-04   | Kharghar    | 10     |
|    3 | Priya  | priya@gmail.com |  60000 | IT         | 2021-07-20   | Kharghar    | 10     |
|    4 | Neha   | neha@gmail.com  |  70000 | Comp       | 2026-09-06   | Pune        | 20     |
|    5 | Nakul  | nakul@gmail.com |  75000 | Comp       | 2026-07-06   | Pune        | 20     |
|    6 | Sai    | sai@gmail.com   |  74800 | AI         | 2026-07-20   | Juhu        | 30     |
|  101 | Rahul  | rahul@gmail.com |  63800 | AI         | 2026-09-18   | Juhu        | 30     |
|  102 | Anjali | anjali@gmail.com |  50000 | Mech       | 2026-06-02   | Ahilyanagar | 40     |
|  103 | Kiran  | kiran@gmail.com |  55000 | Mech       | 2026-07-02   | Ahilyanagar | 40     |
+-------+--------+-----------------+--------+------------+--------------+-------------+--------+
8 rows in set (0.00 sec)
```

18. Write a query to calculate the total salary per department, but only for departments where total salary is greater than 1,00,000 (use HAVING).

```
mysql> select department, sum(salary) as "Total salary" from employee group by
department having sum(salary)>100000;
```

+------------+--------------+

| department | Total salary |

+------------+--------------+

| IT | 120500 |

| Comp | 145000 |

| AI | 138600 |

| Mech | 105000 |

+------------+--------------+

4 rows in set (0.00 sec)

19. Write a query to display all distinct employee names.

mysql> select distinct Name from Employee;

+--------+

| Name |

+--------+

| Ram |

| Priya |

| Neha |

| Nakul |

| Sai |

| Rahul |

| Anjali |

| Kiran |

+--------+

8 rows in set (0.00 sec)

20. Write a query to count the number of departments having the same location.

mysql> select location, count(distinct department) as "Number of Departments"

```
   -> from Employee
   -> group by location;
+-------------+----------------------+
| location    | Number of Departments |
+-------------+----------------------+
| Ahilyanagar |                    1 |
| Juhu        |                    1 |
| Kharghar    |                    1 |
| Pune        |                    1 |
+-------------+----------------------+
4 rows in set (0.00 sec)
```